

ARTIFICIAL INTELLIGENCE

HSU

FEATURE SELECTION

FEATURE SELECTION

- Feature selection is important because it can:
 - Shorten training times
 - Simplify models and make them easier to interpret
 - Enhances testing set performance by reducing overfitting

DROPPING A FEATURE

- Features that can be dropped without impacting the model's precision significantly fall into **2** categories:
- **Redundant**: This is a feature that is highly correlated to other input features and therefore does not add much new information to the signal.
- **Irrelevant**: This is a feature that has a low correlation with the target feature and for that reason provides more noise than signal.

CORRELATION MATRIX – HEAT MAP

A correlation exists between two features when there is a relationship between the different values of the features.

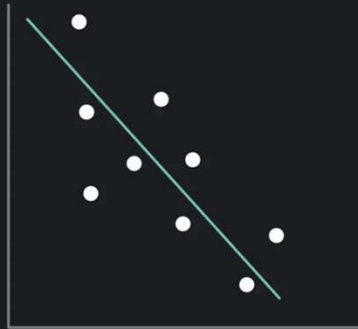
For example, if home prices go up as the square footage goes up, these two features are said to be positively correlated.

If a feature changes consistently in relation to another feature, these features are said to be highly correlated.

Correlation is a continuous value between **-1 and 1**:

- If the correlation is 1, there is a perfect direct correlation.
- If the correlation is -1, a perfect inverse correlation exists.
- If the correlation is 0 between two features, there is no correlation between the two features.

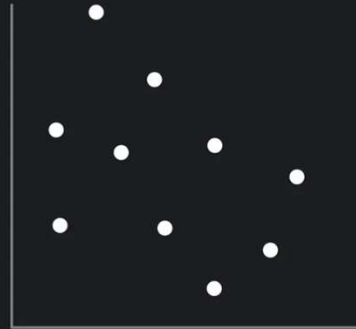
Negative Correlation



The plotted points are clustered around a downward sloping line.

As the values one variable decreases, the other tend to increase and vice versa.

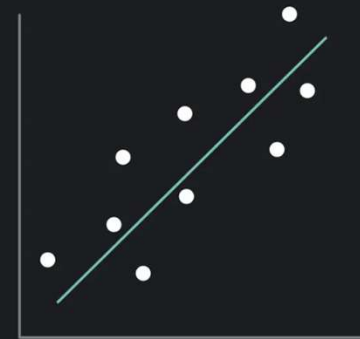
No Correlation



There is no linear trend.

There is no linear correlation between the two variables.

Positive Correlation



The plotted points are clustered around an upward sloping line.

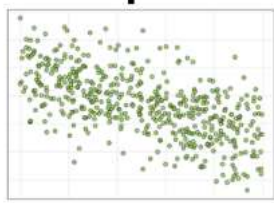
As the values of one variable increase, the value of the other variable tends to increase as well.
As the value of one variable decreases, the value of the other also decreases.

Pearson correlation coefficient r

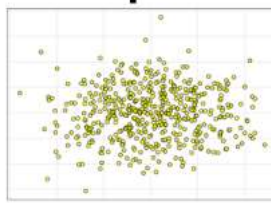
-1 Stronger correlation Weaker correlation **0** Weaker correlation Stronger correlation **1**



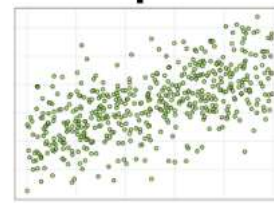
$r = -1$



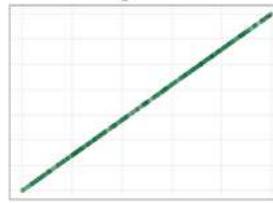
$r = -0.6$



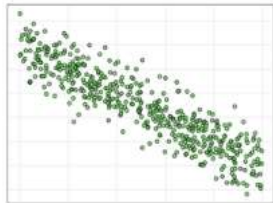
$r = 0$



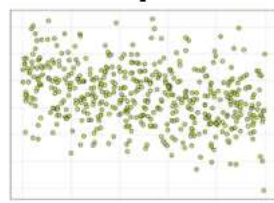
$r = 0.6$



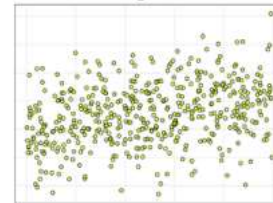
$r = 1$



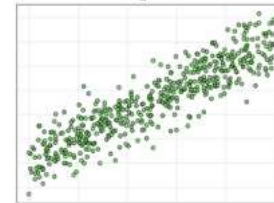
$r = -0.9$



$r = -0.3$



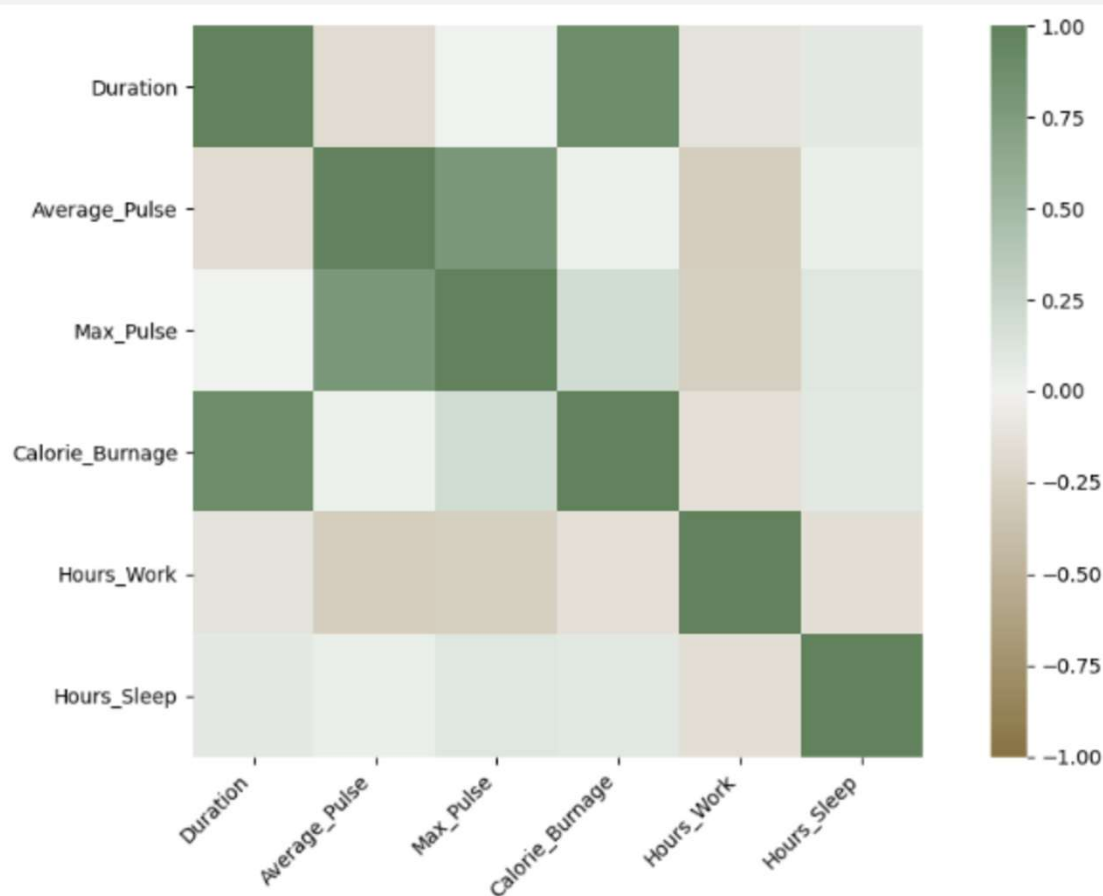
$r = 0.3$



$r = 0.9$

Indirect (negative) correlation

Direct (positive) correlation



The closer the correlation coefficient is to 1, the greener the squares get.

The closer the correlation coefficient is to -1, the browner the squares get.

TS Numpy O ML

	Duration	Average_Pulse	Max_Pulse	Calorie_Burnage	Hours_Work	Hours_Sleep
Duration	1	-0.17	0.00	0.89	-0.12	0.07
Average_Pulse	-0.17	1	0.79	0.02	-0.28	0.03
Max_Pulse	0.00	0.79	1	0.20	-0.27	0.09
Calorie_Burnage	0.89	0.02	0.20	1	-0.14	0.08
Hours_Work	-0.12	-0.28	-0.27	-0.14	1	-0.14
Hours_Sleep	0.07	0.03	0.09	0.08	-0.14	1

- A matrix is an array of numbers arranged in rows and columns.
- A correlation matrix is simply a table showing the correlation coefficients between variables.

WRAPPER-BASED METHODS

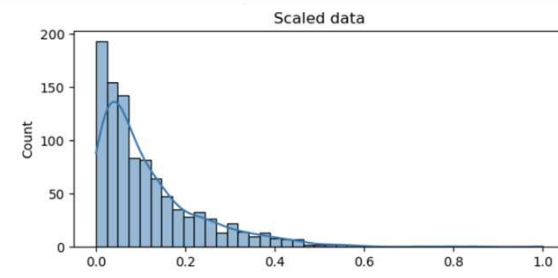
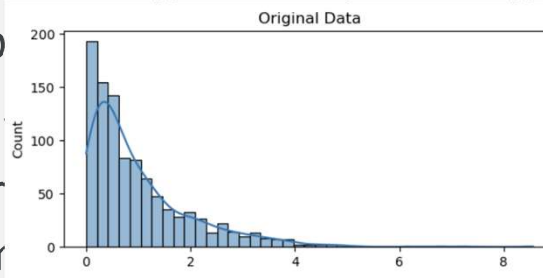
- feature selection is essentially reduced to a search problem using these steps:
- 1. A subset of features is used to train a model
- 2. Based on the results of each iteration, features are added to or removed from the subset

DATA SCALING

- This means that you're transforming your data so that it fits within a specific scale, like 0-100 or 0-1. You want to scale data when you're using methods based on measures of how far apart data points are, like [k-nearest neighbors \(KNN\)](#). With these algorithms, a change of "1" in any numeric feature is given

the same imp

- For example, and US Dollar your prices, n



both Yen
t scale
rice of 1

Yen as important as a difference of 1 US Dollar! This clearly doesn't fit with our intuitions of the world. With currency, you can convert between currencies. But what about if you're looking at something like height and weight? It's not entirely clear how many pounds should equal one inch (or how many kilograms should equal one meter).

DATA NORMALIZATION/SCALING

DATA NORMALIZATION/SCALING

What is Normalization?

- Normalization is a specific form of feature **scaling** that transforms the range of features to a standard scale.
- Normalization is required only when your dataset has features of varying ranges
- Data normalization is the process of reorganizing data within a database so that users can utilize it for further queries and analysis.
- Simply put, it is the process of developing clean data. This includes eliminating redundant and unstructured data and making the data appear similar across all records and fields.

Why Normalize Data?

- Normalized data enhances model performance and improves the accuracy of a model. It aids algorithms that rely on **distance** metrics, such as k-nearest neighbors by preventing features with larger scales from dominating the learning process.
- Normalization fosters stability in the optimization process, promoting faster convergence during gradient-based training. It mitigates issues related to vanishing or exploding gradients, allowing models to reach optimal solutions more efficiently.
- Normalized data is also easy to interpret and thus, easier to understand. When all the features of a dataset are on the same scale, it also becomes easier to identify and visualize the relationships between different features and make meaningful comparisons.

EXAMPLE

- Let's take a simple example to highlight the importance of normalizing data:
 - Square footage: Ranges from 500 to 5000 square feet
 - Number of bedrooms: Ranges from 1 to 5
 - Distance to supermarket: Ranges from 0.1 to 10 miles

EXAMPLE (CONT)

- **Algorithm might give more weight to features with large scales**, such as square footage. During training, the algorithm assumes that a change in total square footage will have a significant impact on housing prices.
- The algorithm might **overlook** the nuances of features that are relatively small, such as the number of bedrooms and distance to the supermarket. This skewed emphasis can lead to suboptimal model performance and biased predictions.
- By normalizing the features, we can ensure that **each feature contributes proportionally** to the model's learning process. The model can now learn patterns across all features more effectively, leading to a more accurate representation of the underlying relationships in the data.

WHY NORMALIZATION?

- We have data of 35 job holders with two variables years of experience ranging from 1.1 years — 10.5 years and their corresponding salaries ranging from \$37k to \$122k.

YearsExperience	Salary
1.1	39343
1.3	46205
1.5	37731
2	43525
2.2	39891
2.9	56642
3	60150
3.2	54445
3.2	64445
3.7	57189
3.9	63218
4	55794
4	56957

First 13 rows of the salary data

WHY NORMALIZATION?

- **Distance-Based algorithms**, such as SVM, K-Means, and KNN classifies objects by finding similarity between them using distance functions. These algorithms are receptive to the size of the variables. If we don't scale, the feature with a higher magnitude (Salary), will have more influence than the feature with a lower magnitude (years of experience), and this leads to bias in data, and it also affects the accuracy.
- **Gradient Descent Based Algorithms** uses the vector-valued function θ . When the scales of the variables are different, θ will descend faster in a lower scale variable as its learning rate will be higher. Therefore, we need to scale the variables to move the gradient with the same learning rate, and this, in turn, will make the convergence faster towards the minima. This can be seen in the diagram below, the left graph is not scaled, and it takes a longer time (as small scale feature has slow learning rate) compared to the scaled data on the right-hand graph.
- **Principal Component Analysis(PCA)** that requires scaling as it tries to capture maximum variance. If the data is not scaled, this could lead to bias as the feature with a larger scale could influence the smaller-scale features.

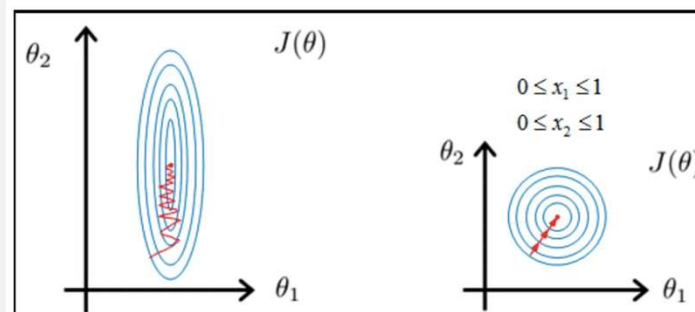


Diagram Credit

NORMALIZATION TECHNIQUES

• Min-Max Scaling

Min-max scaling is very often simply called 'normalization.' It transforms features to between 0 and 1. The formula for min-max scaling is:

$$X_{\text{normalized}} = (X - X_{\min}) / (X_{\max} - X_{\min})$$

where X is a random feature value that is to be normalized.

$X_{\min}$ is the minimum feature value in the dataset, and $X_{\max}$ is the maximum feature value in the dataset.

Min-max scaling is a good choice when:

- The approximate upper and lower bounds of the dataset is known, and the dataset has few or no outliers
- When the data distribution is unknown or non-Gaussian, and the data is approximately uniformly distributed across the range
- When maintaining the distribution's original shape is essential

Normalized Value

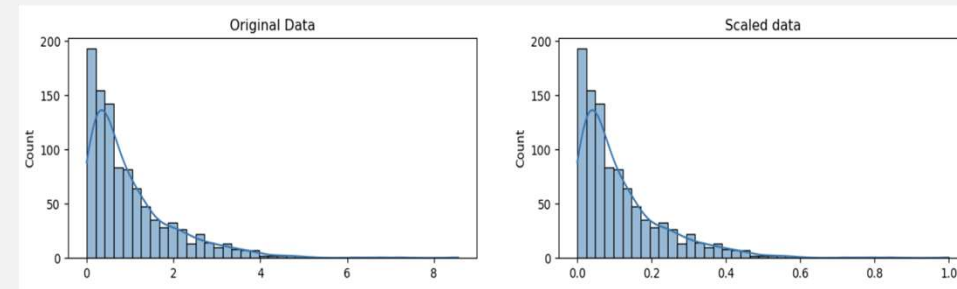
Original Value

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Maximum Value of x

Minimum Value of x

Formula for Min-Max Normalization



- **Z-score normalization (standardization)**

- Z-score normalization (standardization) assumes a Gaussian (bell curve) distribution of the data and transforms features to have a mean (μ) of 0 and a standard deviation (σ) of 1. The formula for standardization is:

$$X_{\text{standardized}} = (X - \mu) / \sigma$$

- This technique is particularly useful when dealing with algorithms that assume normally distributed data, such as many linear models.

The diagram illustrates the formula for standardization: $x' = \frac{x - \mu}{\sigma}$. It includes the following labels and arrows:

- Standardised Value** points to x' .
- Original Value** points to x .
- Sample Mean** points to μ .
- Sample Standard Deviation** points to σ .

Formula for Standardisation

Normalization	Standardization
This technique uses minimum and max values for scaling of model.	This technique uses mean and standard deviation for scaling of model.
It is helpful when features are of different scales.	It is helpful when the mean of a variable is set to 0 and the standard deviation is set to 1.
Scales values ranges between [0, 1] or [-1, 1].	Scale values are not restricted to a specific range.
It got affected by outliers.	It is comparatively less affected by outliers.
Scikit-Learn provides a transformer called MinMaxScaler for Normalization.	Scikit-Learn provides a transformer called StandardScaler for Normalization.
It is also called Scaling normalization.	It is known as Z-score normalization.
It is useful when feature distribution is unknown.	It is useful when feature distribution is normal.

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HOW TO DEAL WITH MISSING DATA?

1. Deletion (rows, columns)

2. Imputation

- Mean, median, mode
- Machine learning based: Linear Regression, KNN, EM, ...

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ONE HOT ENCODING

- One-hot encoding is a technique for representing **categorical data** as numerical vectors, where each unique category is represented by a binary column with a value of 1 indicating its presence and 0 indicating its absence.
- One-hot encoding is a method of converting categorical variables into a format that can be provided to machine learning algorithms to improve prediction.
- It involves creating new binary columns for each unique category in a feature. Each column represents one unique category, and a value of 1 or 0 indicates the presence or absence of that category.

Color		Color	Red	Green	Blue
Red	→	Red	1	0	0
Green		Green	0	1	0
Blue		Blue	0	0	1

- The original "Color" column is replaced by three new binary columns, each representing one of the colors. A value of 1 indicates the presence of the color in that row, while a 0 indicates its absence.

Label Encoding

Food Name	Categorical #	Calories
Apple	1	95
Chicken	2	231
Broccoli	3	50



One Hot Encoding

Apple	Chicken	Broccoli	Calories
1	0	0	95
0	1	0	231
0	0	1	50

ONE HOT ENCODER

Why Use One-Hot Encoding?

It transforms categorical data into a format that machine learning models can easily understand and use. This transformation allows each category to be treated independently without implying any false relationships between them.

Machine learning compatibility

Most machine learning algorithms require numerical input to perform their calculations. Categorical data needs to be transformed into a numerical format for these algorithms to use effectively

Avoiding ordinality

- Label encoding can create problems because it might suggest an order or ranking among categories that doesn't actually exist. For example, assigning 1 to Red, 2 to Green, and 3 to Blue could make the model think that Green is greater than Red and Blue is greater than both.
- This misunderstanding can negatively affect the model's performance. One-hot encoding solves this problem by creating a separate binary column for each category. This way, the model can see that each category is distinct and unrelated to the others.

ADVANTAGES

- The advantages of using one hot encoding include:
 1. It allows the use of categorical variables in models that require **numerical** input.
 2. It can improve model performance by providing **more information** to the model about the categorical variable.
 3. It can help to **avoid the problem of ordinality**, which can occur when a categorical variable has a natural ordering (e.g. “small”, “medium”, “large”).

DISADVANTAGES

- The disadvantages of using one hot encoding include:
 1. It can lead to **increased dimensionality**, as a separate column is created for each category in the variable. This can make the model more complex and slow to train.
 2. It can lead to **sparse data**, as most observations will have a value of 0 in most of the one-hot encoded columns.
 3. It can lead to **overfitting**, especially if there are many categories in the variable and the sample size is relatively small.